

Xianda(Allen) Du

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Education

University of Waterloo — CGPA: 90.87/100

Waterloo, ON

Bachelor of Applied Science in Computer Engineering, Option in Artificial Intelligence

September 2022 – May 2027

Relevant Coursework: Machine Learning, Deep Learning, Reinforcement Learning, Computational Intelligence, Co-operative and Adaptive Algorithms, Algorithms and Data Structures, Systems Programming **Awards:** President's Scholarship of Distinction in 2023; Term Dean's Honour List in Fall 2022, Winter 2023, Fall 2023; Term Distinction in Winter 2025, Spring 2026

Publications

* denotes equal contribution, ⌘ denotes under review, ⚡ denotes in progress.

- [1] Yongan Yu*, Qingchen Hu*, **Xianda Du***, Jiayin Wang, Fengran Mo, and Renée Sieber. *WXImpactBench: A Disruptive Weather Impact Understanding Benchmark for Evaluating Large Language Models*. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 4016–4035, Vienna, Austria. Association for Computational Linguistics.
- [2] Yongan Yu*, **Xianda Du***, Qingchen Hu*, Jiahao Liang*, Jingwei Ni, Dan Qiang, Kaiyu Huang, Grant McKenzie, Renée Sieber, and Fengran Mo. *WeatherArchive-Bench: Benchmarking Retrieval-Augmented Reasoning for Historical Weather Archives*. In *SIGIR 2026*.
- [3] Ahmed Abdelaal, **Xianda Du**, and Amir-Hossein Karimi. *Mean Flows as Controllable Priors for Solving Inverse Problems*. In *NeurIPS 2026 Workshop BeNT0*.
- [4] ⌘ **Xianda Du***, Max Ku*, Weiming Ren, Zhi Rui Tam, Chunlin Ren, Ping Nie, Min-Hung Chen, and Wenhui Chen. *VIEScore2: Unified Image Evaluation with Spatially Grounded Explanations*. In *ICLR 2027*. [Manuscript upon request]
- [5] ⌘ Yuxing Tian*, **Xianda Du***, Weixu Zhang, Jia Ao Sun, Fengran Mo, and Jian-Yun Nie. *Evidence in the Loop: Training Search Agents from Knowledge Graphs*. In *ICLR 2027*. [Manuscript upon request]
- [6] ⌘ Ahmed Abdelaal, **Xianda Du**, and Amir-Hossein Karimi. *Mean Flows as Controllable Priors for Solving Inverse Problems*. In *ICLR 2027*. [Manuscript upon request]
- [7] ⚡ Chunlin Ren, **Xianda Du**, Yuyang Zhang, and Robert Mercer. *What Reward Do the Metrics Imply in Multimodal Sentiment Analysis?* In preparation.

Research Experience

Research Assistant – VIEScore2

University of Waterloo

Advisor: Prof. Wenhui Chen; Mentored by Max Ku

October 2025 – Present

- Co-first authored **VIEScore2**, a unified VLM evaluator for image generation and editing that jointly predicts quality scores and defect locations on a text-native $N \times N$ grid, with a parameter-free parser that turns predictions into spatially grounded explanations.
- Unified **38K** training examples from **5 sources** (ImagenWorld, RichHF, PAL4VST, etc.) with score-only, localization-only, and joint supervision; fine-tuned **Qwen3-VL-8B** via **SFT** (DeepSpeed ZeRO-3) followed by **GRPO** with cell-level **Dice**, score-accuracy, and format rewards.
- Reached overall-score **SRCC 0.601** vs. **0.491** for Gemini-3-Flash, the strongest zero-shot VLM baseline; ranked **1st** in grid IoU on **3 of 6** defect-localization benchmarks and top-3 on 5, ahead of GPT-5.6 and Claude.

Research Assistant – Controllable Generative Priors

University of Waterloo

Advisor: Prof. Amir-Hossein Karimi; Mentored by Ahmed Abdelaal

August 2026 – Present

- Formulated pretrained **MeanFlow** finite-interval transport maps as **controllable priors** for training-free inverse problems, enabling one-evaluation clean-endpoint prediction and running existing flow solvers (**D-Flow**, **PnP-Flow**, **MPC-Flow**) without retraining.
- Introduced **Receding-Horizon State Optimization (RHISO)**, a model-predictive controller over the generative state; best in **35 of 60** task-metric cells and **200 of 240** pairwise comparisons across **5 ImageNet restoration tasks**, **4 solvers**, and **4 priors**.
- Built a unified benchmark over pixel- and latent-space priors (**JiT**, **pMF**, **SiT-XL**, **iMF**) spanning **PyTorch** and **JAX**, with backend-agnostic forward operators, sharded **SLURM** sweeps on Alliance clusters, and PSNR/SSIM/LPIPS reporting.

Research Assistant – Evidence-Grounded Search Agents

Université de Montréal

Advisor: Prof. Jian-Yun Nie; mentored by Yuxing Tian

July 2026 – September 2026

- Co-first authored **Evidence-in-the-Loop**, a self-evolving **Proposer-Solver** framework that trains LLM search agents with **RL** on questions synthesized online from **multi-hop Wikidata KG paths**, keeping only paths whose every triplet is backed by verified, **E5**-retrievable Wikipedia evidence.

- Designed two **process rewards**, **evidence acquisition** and **evidence utilization**, that credit actual retrieval of verified passages instead of entity mentions, curbing the **reward hacking** of entity-based rewards (E-GRPO, KG-Path).
- Achieved the **best average EM/F1 in all 5 training settings** (Qwen2.5-7B Base/Instruct, LLaMA-3.1-8B, Qwen3-8B, Search-R1-7B) over SSP and KG-Path, winning **18 of 20** multi-hop comparisons; Proposer-generated questions lift downstream multi-hop F1 from **30.4 to 35.3**.

Research Assistant – LLM Reranking (RankLLM)

University of Waterloo

Advisor: Prof. Jimmy Lin

June 2026 – Present

- Contributing to **RankLLM**, Castorini's open-source toolkit for **LLM-based listwise reranking**: integrated **SPLADE-v3** learned sparse retrieval as a first-stage retriever and upgraded the **Pyserini** backend to v2.4.0.
- Improved reranking reproducibility by enforcing deterministic decoding, and wrote a Colab CLI onboarding guide covering installation, sanity checks, and end-to-end **RankZephyr** reranking.

Research Assistant – Weather Archival Rescue

McGill University

Advisor: Prof. Renée Sieber; in collaboration with RBC Borealis AI and CEIMIA

September 2024 – Present

- **WXImpactBench (ACL 2025 Findings, co-first author)**: built the first benchmark for LLM understanding of disruptive weather impacts (**1,736 samples**, **6 impact categories**) with multi-label classification and **100-passage ranking QA**; evaluated **8+ LLMs** (Llama-3, Qwen2.5, Mixtral, GPT) with 4-bit quantization.
- **WeatherArchive-Bench (SIGIR 2026, co-first author)**: built a retrieval-augmented reasoning benchmark over **1M+ archival segments** and **335 expert queries** with **IPCC-aligned** vulnerability assessment; benchmarked **13+ retrievers** (BM25, SPLADE, ANCE, dense, cross-encoder reranking) and **17 LLMs**, peaking at **95.8% Recall@100**.
- **WXChat**: developing a digitization pipeline for Victorian-era newspapers with **Tesseract 5** OCR at 600 DPI, layout-aware column segmentation (**OpenCV**; skew correction, CLAHE, Sauvola binarization), and async **GPT-4o-mini** post-OCR correction.

Undergraduate Research Assistant – Noise Injection Analysis

University of Waterloo

Advisor: Prof. En-Hui Yang

January 2025 – April 2025

- Studied effects of **Gaussian noise** (9 levels, $\sigma=0.1-0.9$) and **FGSM adversarial perturbations** on intermediate feature representations in pretrained **ResNet-18** on **ImageNet**.
- Extracted and analyzed feature maps across **4 network layers** (56×56 to 7×7) using **PyTorch** forward hooks, generating statistical summaries and 2D cross-layer distribution histograms.

Professional Experience

Autonomy and Algorithm Engineer Intern

September 2026 – December 2026

Waabi

Toronto, ON

- Working on **novel view synthesis (NVS) quality evaluation** for sensor-simulation reconstruction models in self-driving, an exploratory effort to design metrics and pipelines that assess rendering fidelity at novel viewpoints.

Machine Learning Engineer

January 2026 – April 2026

StackAdapt

Toronto, ON

- Shipped StackAdapt's **Precision Audience** LLM pipeline that converts unstructured advertiser ICPs into executable boolean segment plans across **9 audience sources** $\times$ **10 geo locations** per request, built on **GPT-5** with **Pydantic**-typed structured outputs and an **async** worker pool backed by **Redis** job tracking.
- Eliminated silent data corruption by building a **4-layer hallucination detection** system (schema enforcement, **reference-grounding** against candidate sets, strict enum validation, and post-LLM **safeguard rules**), ensuring every audience segment in the output corresponds to a validated upstream source.
- Reduced unsafe ad targeting and improved reliability by designing **prompt hardening** (injection detection, 10-category sensitive-data taxonomy) and a typed **exception hierarchy** that maps failures to correct **HTTP semantics**, enabling per-geo **partial-success (207)** so multi-country requests degrade gracefully.
- Scaled audience coverage to **~12.7K segments** across multiple countries by building **hybrid vector retrieval** on **Qdrant** (**dense + BM25 + RRF**, geo-indexed prefilter) and an **end-to-end data pipeline** with ISO validation, diff-based incremental ingestion, and **snapshot/restore** rollback to S3.

WatAI RAG Project Technical Lead

April 2025 – December 2025

University of Waterloo, WatAI

Waterloo, ON

- Designed and implemented a research-oriented **RAG** system for educational QA, supporting **Qwen**, **OpenAI**, and **Gemini** through a flexible **LangChain**-based interface with configurable prompts and streaming inference.
- Built an end-to-end retrieval pipeline over course PDFs and slides with document chunking, dense retrieval, optional **hybrid BM25+dense** search, **re-ranking**, and citation-grounded generation.
- Developed evaluation and monitoring workflows using **Recall@k**, **nDCG**, and **LLM-as-a-judge** faithfulness checks to study retrieval quality and response grounding in real-user interactions.
- Implemented tool-augmented querying and model routing strategies to compare trade-offs across model quality, latency, and cost in practical RAG settings.

Platform Developer

September 2024 – December 2024

University of Waterloo, ECE Department

Waterloo, ON

- Fine-tuned **Hugging Face** sentiment analysis models in Python using **TensorFlow/PyTorch** and analyzed model behavior for media content classification, achieving **80%** accuracy.
- Used **pandas** to curate and label collected data, and built a pipeline that transformed embeddings and metadata from a **PostgreSQL** database on **GKE** into a self-designed **neural network** for scraped article classification.
- Strengthened experimental workflow design across data preprocessing, representation learning, and downstream classification, with emphasis on reproducible ML development rather than product-facing deployment.

Web 3.0 Quantitative Trading Challenge

Jun 2025 – Nov 2025

- Framed 15-min crypto returns as **ranking** and **Weighted Spearman**; trained **TabNet**, **XGBoost**, **MLP**, and **LightGBM** via **Optuna**; finalized **Optuna LightGBM v10**, scoring **0.57 Weighted Spearman** on the public leaderboard.
- Implemented **leak-safe**, **rolling-window** features (RSI divergence, distance-to-high/low, volatility) via vectorized **NumPy/pandas**; importance screening kept **80** features, improving generalization/training speed.
- Built **8-model LightGBM ensemble** (varying depth/regularization/seeds/losses) with **weighted blending**; applied adaptive clipping/winsorization, normalization, smoothing to stabilize ranks and beat single models.

Skills

Languages: Chinese (Native), English (Fluent)

Programming Languages: Python, JavaScript/TypeScript, C/C++, Java, SQL, Bash, R, MATLAB, HTML/CSS

Developer Tools: Git, Docker, Kubernetes, Cloud Platforms, Web Development, Scrapy, Supabase, Qdrant

ML/AI Libraries: PyTorch, TensorFlow, Keras, Transformers, Scikit-learn, NumPy, Pandas, OpenCV, Matplotlib

Frameworks/Areas: RAG, Vision-Language Models, Information Retrieval, CNN, LSTM, XGBoost, LightGBM, HPO

COMMUNITY SERVICE

SIGIR 2026 Resource Track Reviewer

MeLLM @ ACL 2026 Reviewer

Developer of open source toolkit RankLLM